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In [4]: import numpy as np
import pandas as pd
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
import matplotlib.pyplot as plt
```

```
In [5]: data = pd.read_csv("data/diabetes.csv")
print(data.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
In [6]: X = data.drop("Outcome", axis = 1)
Y = data["Outcome"]

cols = ["Glucose", "BloodPressure", "Insulin", "BMI"]
for col in cols:
    X[col] = X[col].replace(0, X[col].mean())
```

```
In [7]: x_train, x_test, y_train, y_test = train_test_split(X, Y, test_size = 0.2, random_state = 42)
rf = RandomForestClassifier(
    n_estimators = 500,
    criterion = "entropy",
    class_weight = "balanced",
    random_state = 42)
rf.fit(x_train, y_train)
```

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Out[7]: ▼ RandomForestClassifier ⓘ ?
        ► Parameters
```

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In [8]: y_pred = rf.predict(x_test)
```

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In [9]: print("Accuracy Score:", accuracy_score(y_test, y_pred))
print("Confusion Matric:", confusion_matrix(y_test, y_pred))
print("Classification report:", classification_report(y_test, y_pred))
```

Accuracy Score: 0.7662337662337663

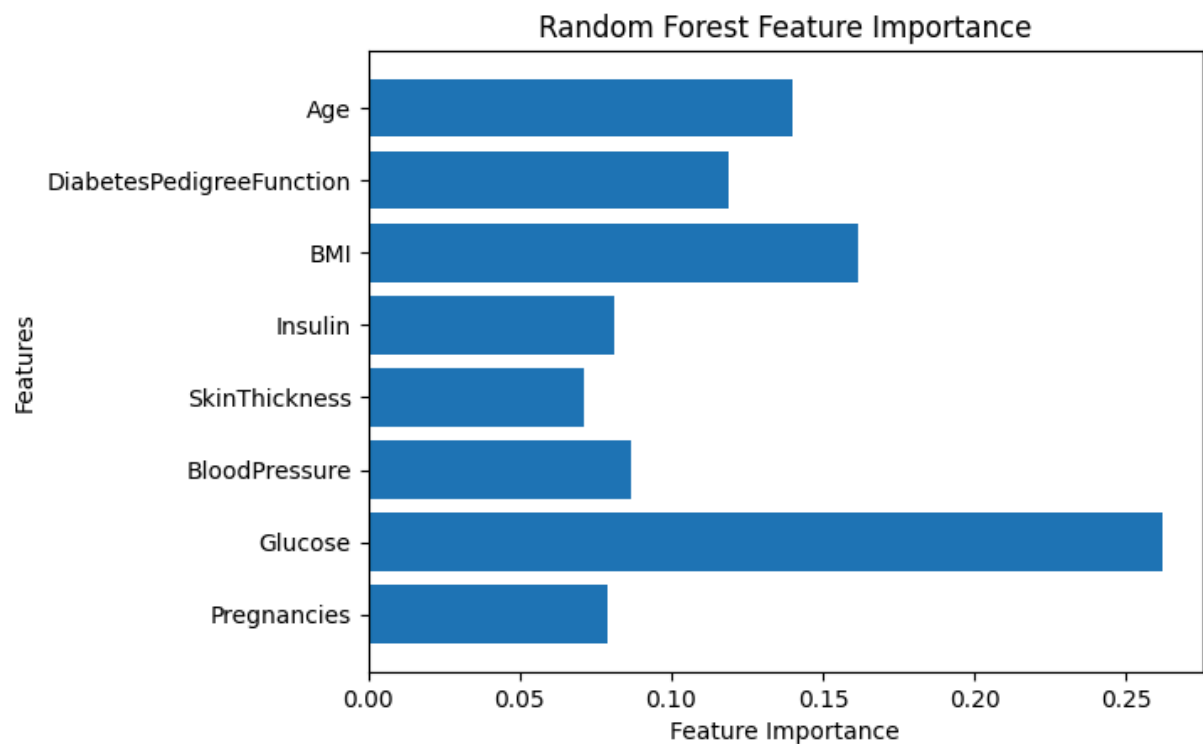
Confusion Matric: [[81 18]

[18 37]]

Classification report:

			precision	recall	f1-score	support
	0	0.82	0.82	0.82		99
	1	0.67	0.67	0.67		55
	accuracy			0.77		154
	macro avg	0.75	0.75	0.75		154
	weighted avg	0.77	0.77	0.77		154

```
In [9]: importance = rf.feature_importances_  
features = X.columns  
  
plt.barh(features, importance)  
plt.xlabel("Feature Importance")  
plt.ylabel("Features")  
plt.title("Random Forest Feature Importance")  
plt.show()
```



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In [ ]:
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